MIL-T-82535(OS)
18 April 1968

Superseding
NOTS XS 546 Rev. C
29 September 1967

MILITARY SPECIFICATION
FOR
TIMER, INTERVAL
MARK 11 MOD 2
(ASSEMBLY)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements of one type of Interval Timer Mark 11 Mod 2 Assembly. (See 6.1 for intended use and description.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein, except as noted.

SPECIFICATIONS

FEDERAL

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

MILITARY

MIL-P-116	Preservation, Methods of
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

FSC 1340

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STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels

MILITARY

MIL-STD-105 D	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 540655	Tester, Timer, T-3077 (Assembly)
LD 541525	Timer, Interval, Mk 11 Mod 2, Assembly and all documents listed thereon

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

OD 12996	Operation and Maintenance of the Interval Timer Test Set T-3077

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the timer shall be in accordance with the requirements specified herein and with the applicable documents listed in BuWeps LD 541525.

3.1.1 Electrical component requirements. Electrical components used in the manufacture of the timer shall comply with the requirements of their respective drawings and shall be tested as specified in Section 4 (see 4.2).

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3.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the timer shall be manufactured using the methods and procedures proposed for production. The sample shall be tested as specified in Section 4 herein, and is for the purpose of determining that, prior to starting production, the contractor' s production methods are capable of yielding items that comply with the technical requirements of the contract.

3.3 Performance and product characteristics. The timer shall meet the following performance and product characteristics.

3.3.1 Time intervals. With 24 to 32 volts DC applied, the timer shall produce electrical output pulses at two separate, predetermined time intervals (measured from the instant of voltage application) while stabilized at the following temperatures:

a. Plus 35 plus 5 minus 0 degrees F, for not less than 4 hours
b. Plus 80 plus or minus 15 degrees F, for not less than 4 hours
c. Plus 120 plus 0 minus 5 degrees F, for not less than 4 hours

3.3.1.1 Timer interval, 12.5 seconds. Before experiencing any test of 4.5, the timer shall produce an electrical output pulse at 12.5 plus or minus 0.60 seconds. After experiencing any test of 4.5, the timer shall produce an electrical output pulse at 12.5 plus or minus 1.0 seconds. The 12.5-second time interval shall remain within the specified limits when random noise as indicated in Figure 1 appears on the DC input. The noise voltage spikes shall have a duration of 15 plus or minus 5 microseconds, a frequency of 1000 plus or minus 100 cycles per second (cps), and an amplitude of 17 volts above or below the input DC level. Duration of the noise voltage spikes shall be measured at 0.10 V_S above or below the input DC level.

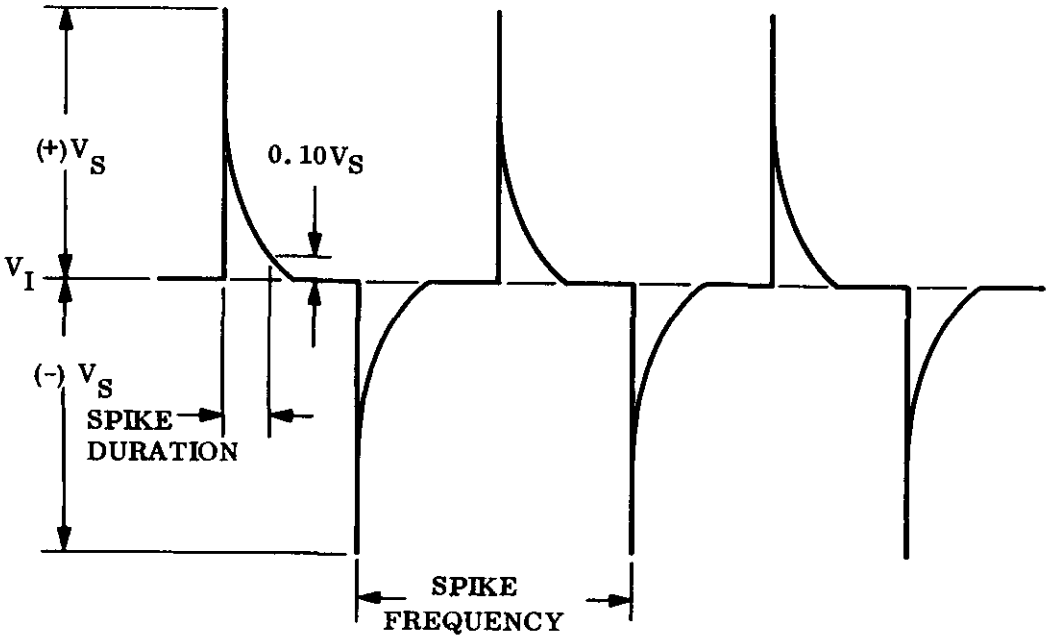
3.3.1.2 Timer interval, 18.5 second. Before experiencing any test specified in 4.5, the timer shall produce an electrical output pulse at 18.5 plus or minus 0.90 seconds. After experiencing any test of 4.5, the timer shall produce an electrical output pulse at 18.5 plus or minus 1.5 seconds.

3.3.2 Energy output. The energy output of the electrical pulses of 3.3.1 shall be not less than 10,500 ergs through a resistance of 5 ohms during the first millisecond, with a peak amplitude of 5 volts DC minimum.

3.3.3 Premature output pulse. Interruption of input voltages shall not cause the 18.5-second interval section of the timer to produce the output pulse specified in 3.3.1.2 before the predetermined time interval has elapsed.

3.3.4 Leakage current. The leakage current between the output leads and the ground leads of the 12.5-second and the 18.5-second time interval sections of the timer, measured through a resistance of 5 ohms, shall be not more than 5 milliamperes prior to the electrical output pulse.

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V_I = INPUT D.C. VOLTAGE (24 TO 32 VOLTS)

V_S = AMPLITUDE OF NOISE VOLTAGE SPIKES (POSITIVE AND NEGATIVE WITH RESPECT TO INPUT VOLTAGE LEVEL V_I)

Figure 1. Random Noise Voltage Spikes

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3.4 Environmental requirements. Unless otherwise specified, the timer shall be capable of conforming to the performance and product characteristics of this specification after being exposed to the environments and all conditions specified herein.

3.4.1 Temperature.

3.4.1.1 Operating condition. The timer shall meet the performance and product characteristics while at temperatures from 35 to 120 degrees F.

3.4.1.2 Storage condition. The timer shall withstand temperatures from minus 65 to plus 160 degrees F.

3.4.2 Vibration.

3.4.2.1 Standby condition. The timer shall withstand vibration as specified for Type I equipment in MIL-STD-167 up to 25 cps.

3.4.2.2 Operating condition. The timer shall withstand vibration through a frequency range of 20 to 2000 cps at 0.10-inch double amplitude or 10g (g = unit of gravitational acceleration) whichever is less. The timer shall meet the requirement of 3.3.1.1 and 3.3.1.2 while being subjected to this environment and thereafter shall meet the performance and product characteristics of this specification.

3.4.3 Shock.

3.4.3.1 Standby condition. The timer shall withstand shocks of 40g peak magnitude. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 5 milliseconds.

3.4.3.2 Operating condition.

3.4.3.2.1 Major. The timer shall withstand a shock of 2400g peak magnitude. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 1.00 plus or minus 0.25 millisecond. The shock shall be applied in the direction shown in Figure 2 when the 18.5-second timer section has operated 9 plus or minus 3 seconds. The 18.5-second timer section shall meet the requirement of 3.3.1.2 while being subjected to this shock, and the 12.5-second timer section shall meet the requirements of 3.3.1.1 immediately after the shock. The timer shall thereafter meet the performance and product characteristics of this specification.

3.4.3.2.2 Minor. The timer shall withstand two shocks of 1200g peak magnitude each. The shocks shall have a pulse shape approximating a one-half sine wave having a duration of 1.0 plus or minus 0.25 millisecond. The shocks shall be applied once in each direction, as shown in Figure 2, when the 18.5-second timer section has operated 9 plus or minus 3 seconds. The 18.5-second

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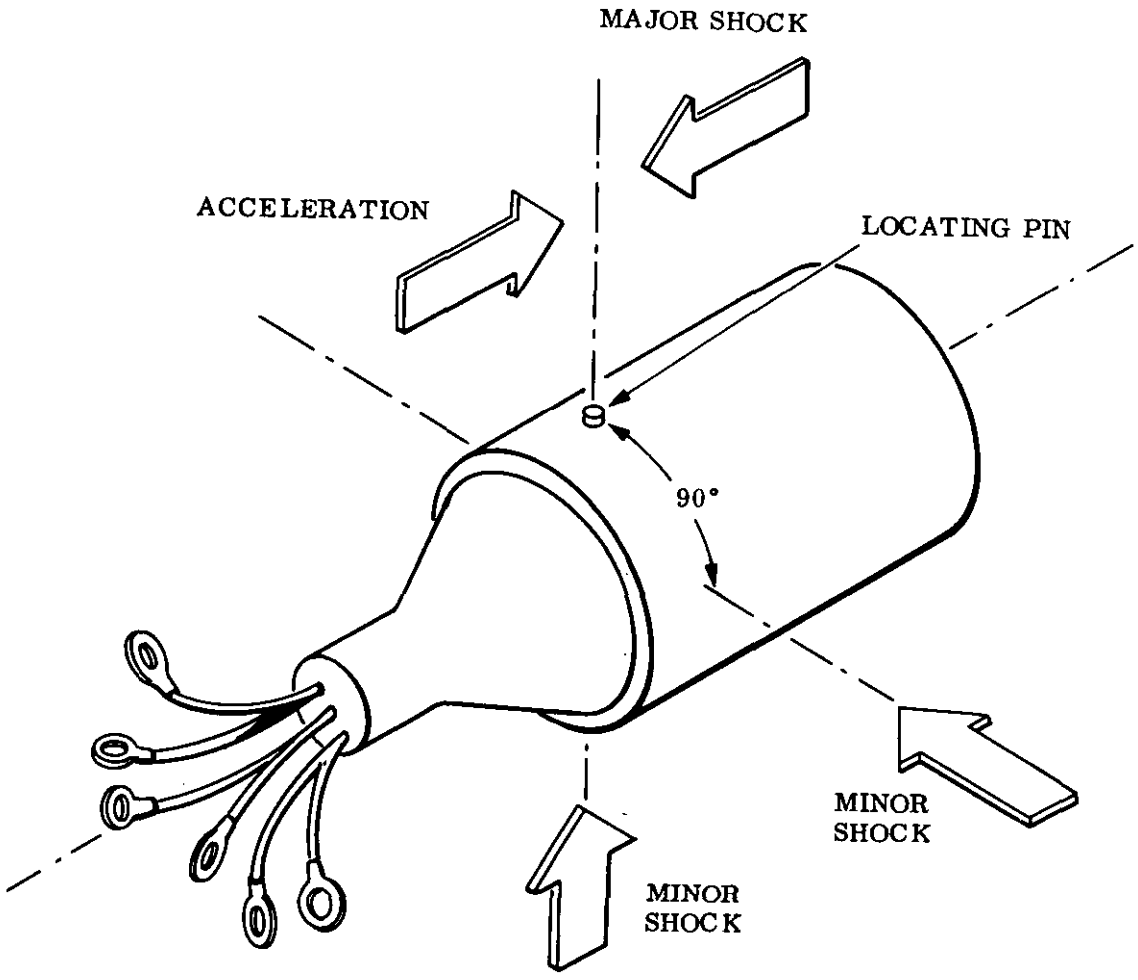


Figure 2. Environmental Testing Axes

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timer section shall meet the requirements of 3.3.1.2 while being subjected to these shocks, and the 12.5-second timer section shall meet the requirements of 3.3.1.1 immediately after these shocks. The timer shall thereafter meet the performance and product characteristics of this specification.

3.4.4 Acceleration. The timer shall withstand an acceleration of 25g for 4 seconds minimum, applied in the direction shown in Figure 2.

3.5 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government' s notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the timer shall be classified as follows:

a. Preproduction and periodic production inspection. (See 4.5.)
b. Quality conformance inspection. (See 4.6.)
c. Electrical component inspection (See 4.7 and 6.3.2.)

4.3 Sampling. (See 6.2.1.)

4.3.1 Lot size. The term "lot" shall mean "inspection lot", that is, a collection of product submitted by a supplier for inspection. Unless otherwise specified in the contract or order, the number of product in each lot shall be three hundred (300).

4.3.2 Preproduction sample. A preproduction sample of fifteen (15) timers, manufactured in accordance with 3.2, shall be subjected to the preproduction tests detailed in 4.5. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the timer by the supplier, prior to approval of the preproduction sample, shall be at the supplier' s risk.

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4.3.3 Periodic production sample. From each succeeding lot of three hundred (300) or less timers, a random sample of ten (10) timers shall be selected. These sample units shall be subjected to the tests of 4.5. The timers shall be considered satisfactory if they meet all specification and pertinent drawing requirements and withstand the tests specified in 4.5. (See Table I.) When a defective unit is found in any lot, acceptance of all units in the lot shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary action, with the concurrence of the procuring activity, to correct or eliminate the defect from all units in the lot and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

4.3.4 Quality conformance inspection sampling. Samples shall be selected as specified in 4.6 and shall be used to determine the acceptability of the timer inspected as specified in 4.6.

Table I. Preproduction and Periodic Production Test Sequence

Test	Paragraph Number	Test Quantities Preproduction	Periodic
Quality Conformance Inspection	4.6	15	10
Temperature, Storage Condition	4.5.1.1	15	10
Quality Conformance Inspection	4.6*	15	10
Vibration, Standby Condition	4.5.1.2	15	10
Quality Conformance Inspection	4.6*	15	10
Shock, Standby Condition	4.5.1.3	15	10

*Time interval tests of 4.6.1.1 shall be conducted only at plus 80 plus or minus 15 degrees F.

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Table I. Preproduction and Periodic Production Test Sequence (Continued)

Test	Paragraph Number	Test Quantities Preproduction	Periodic
Quality Conformance Inspection	4.6*	15	10
Acceleration	4.5.1.4	15	10
Vibration, Operating Condition	4.5.1.5**	15	10
Quality Conformance Inspection	4.6*	15	10
Shock, Operating Condition	4.5.1.6***	15	10
Quality Conformance Inspection	4.6*	15	10
Temperature, Operating Condition	4.5.1.7	15	10
Low Extreme	4.5.1.7.1	15	10
Quality Conformance Inspection	4.6****	15	10
High Extreme	4.5.1.7.2	15	10
Quality Conformance Inspection	4.6*****	15	10

*Time interval test of 4.6.1.1 shall be conducted only at plus 80 plus or minus 15 degrees F.

**Time interval tests 4.6.1.1.1 and 4.6.1.1.2 shall be performed during environmental tests; thereafter, the remaining quality conformance inspection test shall be performed.

***Time interval test 4.6.1.1.2 shall be performed during the environmental test; thereafter, the remaining quality conformance inspection tests shall be performed. (See 4.5.1.6.1 and 4.5.1.6.2.)

****Perform quality conformance inspection 4.6 only at plus 35 plus 5 minus 0 degrees F.

*****Perform quality conformance inspection 4.6 only at plus 120 plus 0 minus 5 degrees F.

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4.4 Test equipment and test conditions.

4.4.1 Quality conformance inspection test equipment. The following items of test equipment are required to perform the quality conformance inspection:

	Equipment	Capability
a.	Tester, Timer T-3077 (Assembly)	Conform to LD 540655 Operate in accordance with OD 12996
b.	Temperature chamber	Maintain temperature at: Plus 35 plus 5 minus 0 degrees F Plus 80 plus or minus 15 degrees F Plus 120 plus 0 minus 5 degrees F

4.4.2 Preproduction and periodic production test equipment. The following items of test equipment, in addition to selected items of quality conformance inspection equipment and pieces of equipment recommended by applicable specifications or standards referenced within test procedure paragraphs, are required to perform the preproduction and periodic production tests:

	Equipment	Capability
a.	Temperature chamber	Maintain temperature at: Minus 65 plus 5 minus 0 degrees F Plus 160 plus 0 minus 5 degrees F
b.	Vibration testing machine	Control frequency from 20 cps to 2000 cps while subjecting the test item to 0.10-inch double amplitude or 10g acceleration, whichever is less; maintain direction of vibration along specified axes.
c.	Centrifuge	0 to 27.5g; maintain acceleration at 25 plus 2.5 minus 0g.
d.	Shock equipment,* standby	Apply shocks of 40 plus or minus 4g; a one-half sine wave pulse shape having a duration of 5 plus 3 minus 0 milliseconds; maintain direction along specified axes.

*Instrumentation used in conjunction with the shock tester shall not contain a filter having a cutoff frequency below 10 kilocycles.

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e.	Shock equipment,* operating	Apply shocks of up to 2400g plus or minus 10 percent; a one-half sine wave pulse shape having a time duration of 1 plus or minus 0.25 millisecond; maintain direction along specified axes.

*Instrumentation used in conjunction with the shock tester shall not contain a filter having a cutoff frequency below 10 kilocycles.

4.4.3 Electrical component inspection test equipment. The following items of test equipment are required to perform the electrical component inspection shock tests:

	Equipment	Capability
a.	Shock equipment*	Apply shocks of up to 2400g plus or minus 10 percent; a one-half sine wave pulse shape having a duration of 1 plus or minus 0.25 millisecond; maintain direction along specified axes.
b.	Shock equipment *#	Apply shocks of up to 2430 plus or minus 110g; a one-half sine wave pulse shape having a duration of 1 plus or minus 0.25 millisecond; maintain direction along specified axes.
c.	Temperature chambers	Maintain temperature at: Minus 65 plus 5 minus 0 degrees F Plus 160 plus 0 minus 5 degrees F

*Instrumentation used in conjunction with the shock tester shall not contain a filter having a cutoff frequency below 10 kilocycles. The monitoring equipment used for monitoring the components during shock shall be capable of detecting shorts or open circuits of one microsecond duration or less.

#Required for resubmittal shock test only (see 4.7.1.3)

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4.4.4 Test conditions. Unless otherwise specified, the timer shall be subjected to the preproduction and periodic production tests, quality conformance inspection tests, and electrical component inspection tests under the following conditions:

a. Temperature	Room ambient (65 to 95 degrees F)
b. Altitude	Normal ground atmospheric pressure
c. Humidity (Relative)	Room ambient up to 95 percent
d. Line voltage isolation*	The tester specified in 4.4.1a shall be connected to a voltage source that is electrically isolated from all other electrically operated devices. The temperature chamber specified in 4.4.1b shall be turned off during the quality conformance inspection tests at 35 and 120 degrees F.

*Applicable only to quality conformance inspection tests

4.5 Preproduction and periodic production inspection tests. The preproduction and periodic production samples, after satisfactorily passing the quality conformance tests detailed in 4.6 shall be subjected to the tests specified in 4.5.1.

4.5.1 Test procedure. Subject the timer to the test sequence listed in Table I. Unless otherwise specified in the individual test paragraph, the quality conformance inspection, as specified in Table I shall be performed subsequent to environmental tests under conditions specified in 4.4.4. Failure of the timer to pass any of these tests shall constitute a defective unit.

4.5.1.1 Temperature, storage condition. The timer shall be subjected to a temperature of minus 65 plus 5 minus 0 degrees F for not less than 48 hours. The temperature shall be raised to plus 160 plus 0 minus 5 degrees F in not less than 2 hours and maintained at this temperature for not less than 48 hours. The unit shall then be subjected to room temperature for not less than 4 hours prior to performing subsequent tests.

4.5.1.2 Vibration, standby condition. The timer, rigidly mounted as shown in Figure 3, shall be subjected to the vibration tests for Type I equipment as specified in MIL-STD-167 up to 25 cps in each of the three axes shown in Figure 2.

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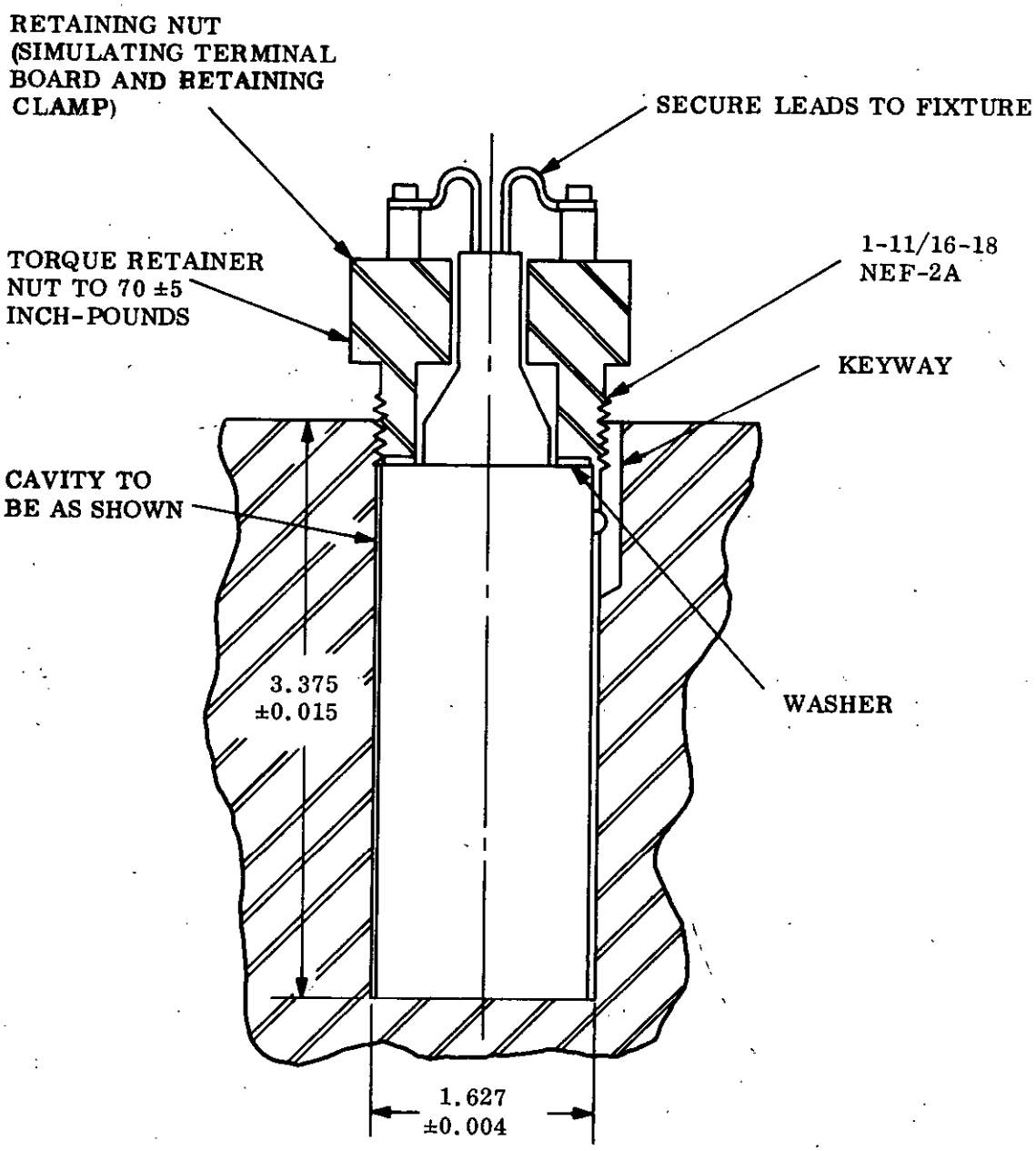


Figure 3. Test Fixture for Preproduction and Product Verification Tests

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4.5.1.3 Shock, standby condition. Subject the timer to shocks of 40 plus or minus 4g peak magnitude with a pulse shape approximating a one-half sine wave, and a duration of 5 plus 3 minus 0 milliseconds. Shocks shall be applied three times in each direction along each of three mutually perpendicular axes as shown in Figure 2. Mount the timer in a manner similar to that shown in Figure 3.

4.5.1.4 Acceleration. The timer, mounted as shown in Figure 3, shall be subjected to acceleration of 25 plus 2.5 minus 0g in the direction shown in Figure 2 for not less than 4 seconds.

4.5.1.5 Vibration, operating condition. The timer, rigidly mounted as shown in Figure 3, shall be subjected to vibration from 20 to 2000 cps at 0.10-inch double amplitude or 10g, whichever is less, along each of three mutually perpendicular axes. The frequency shall be increased uniformly from 20 to 200 cps. The frequency shall be traversed in not less than 2.5 minutes, once in each axis. The timer shall pass the tests of 4.6.1.1.1 and 4.6.1.1.2 while being subjected to this environment and shall meet the remaining quality conformance tests subsequent to this environment.

4.5.1.6 Shock, operating. The timer shall be mounted as shown in Figure 3, and shall be subjected to the following tests:

4.6.1.6.1 Major. Subject the timer to a shock of 2400 plus or minus 240g peak magnitude. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 1 plus or minus 0.25 millisecond. Apply the shock in the direction shown in Figure 2 when the 18.5-second timer section has operated 9 plus or minus 3 seconds. The 18.5-second timer section shall pass the test of 4.6.1.1.2 while being subjected to the shock. Immediately after the shock, the 12.5-second timer section shall be operated and shall pass the test of 4.6.1.1.1. Thereafter the timer shall pass the remaining quality conformance inspection tests of this specification.

4.5.1.6.2 Minor. Subject the timer to two shocks of 1200 plus or minus 120g peak magnitude each. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 1.0 plus or minus 0.25 millisecond. The shocks shall be applied in two directions, as shown in Figure 2, when the 18.5-second timer section has operated 9 plus or minus 3 seconds. The 18.5-second timer section shall pass the test requirement of 4.6.1.1.2 while being subjected to these shocks. Immediately after the shocks the 12.5-second timer section shall be operated and shall pass the test requirements of 4.6.1.1.1. Thereafter the timer shall pass the remaining quality conformance inspection tests of this specification.

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4.5.1.7 Temperature, operating.

4.5.1.7.1 Low extreme. Subject the timer to a temperature of 35 plus 5 minus 0 degrees F for not less than 4 hours, and while at that temperature, the timer shall pass the quality conformance inspection tests.

4.5.1.7.2 High extreme. Subject the timer to a temperature of 120 plus 0 minus 5 degrees F for not less than 4 hours, and while at that temperature, the timer shall pass the quality conformance inspection tests.

4.6 Quality conformance inspection. Samples selected using the plans and procedures set forth in Table II of this specification shall be subjected to the test and examinations as described in 4.6.1.

Table II. Quality Conformance Tests and Examinations, and AQL

Test	Sampling Plan	Test Paragraph	AQL
Timer interval, 12.5-second	*	4.6.1.1.1	*
Timer interval, 18.5-second	*	4.6.1.1.2	*
Energy output, 12.5-second	**	4.6.1.2	1.0
Energy output, 18.5-second	**	4.6.1.2	1.0
Premature output pulse	*	4.6.1.3	*
Leakage current, 12.5-second section	*	4.6.1.4.1	*
Leakage current, 18.5-second section	*	4.6.1.4.2	*
Visual examination	*	4.6.1.5	*
Preservation, packaging, packing and marking	*	4.8	*

*One-hundred percent inspection; defect is cause for rejection of individual unit.
**MIL-STD-105 D Table Inspection Level II.

4.6.1 Test procedure. The timer shall be subjected to the tests and examinations as outlined herein. Unless otherwise specified, inability of the timer to pass the tests and examinations shall constitute a failure.

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4.6.1.1 Timer intervals. The timer shall be subjected to the following tests while stabilized at the following temperatures:

a. Plus 35 plus 5 minus 0 degrees F, for not less than 4 hours

b. Plus 80 plus or minus 15 degrees F, for not less than 4 hours

c. Plus 120 plus 0 minus 5 degrees F, for not less than 4 hours

4.6.1.1.1 Timer interval, 12.5-second. Using the tester specified in 4.4.1a, apply a potential of 24 volts DC and random noise, as described in 3.3.1.1 and Figure 1, to the 12.5-second section input. Record the time interval from the instant voltage is applied until the electrical output pulse is produced. Repeat the test at 28 and 32 volts DC, allowing a minimum of 2 minutes between successive tests. The time intervals for each of these tests shall be 12.50 plus or minus 0.60 seconds; sample timers, which have been subjected to any of the tests of 4.5.1, shall have time intervals of 12.50 plus or minus 1.0 seconds. During the test at 24 volts DC, observe and record the electrical output pulse of the timer, using the tester oscilloscope camera.

4.6.1.1.2 Timer interval, 18.5-second. Apply a potential of 24 volts DC, using the tester specified in 4.4.1a, to the 18.5-second section input of the timer. Record the time interval period until the electrical output pulse is produced. Repeat this test at 28 and 32 volts DC, allowing a minimum of 2 minutes between successive tests. The time intervals for each of these tests shall be 18.50 plus or minus 0.90 seconds; sample timers, which have been subjected to any of the tests of 4.5.1, shall have time intervals of 18.5 plus or minus 1.5 seconds. During the test at 24 volts DC, observe and record the electrical output pulse of the timer, using the oscilloscope camera.

4.6.1.2 Energy output. Compute, in accordance with the procedure outlined in 4.6.1.2.1, the energy output of the electrical output pulses of 4.6.1.1.1 and 4.6.1.1.2. The energy output shall be not less than 10,500 ergs through a resistance of 5 ohms during the first millisecond with a peak amplitude of 5 volts DC minimum.

4.6.1.2.1 Energy output calculation procedure. From the photographic record of the output pulse of each timer section, measure the output voltage at 0.25 and 0.75 millisecond from the beginning of the pulse and identify these measurements as E_1 and E_2 respectively (see Figure 4). The energy output of the pulse is defined here as the summation of energy increments calculated from the two voltages and the two one-half millisecond time intervals. The basic energy equation is:

$$\text{Energy (ergs)} = \frac{E^2}{R_L} \times T \times 10^7$$

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Where: E = voltages E_1, E_2 (volts)

R_L = load resistance (ohms)

T = time interval (seconds)

10^7 = conversion factor from watt-seconds or joules to ergs

In this test, R_L = 5.00 ohms, and T = 0.0005 second. The energy output of the pulse shall then be calculated from the following:

Energy (ergs) = $(E_1^2 + E_2^2) \times 10^3$

A typical output pulse is shown in Figure 4.

4.6.1.3 Premature output pulse. Apply a potential of 32 volts DC to the 18.5-second section input, using the tester specified in 4.4.1a. When the timer has operated for a period of 5 plus or minus 1 second, remove the applied potential and observe and verify that an electrical output pulse exceeding 1.5 volts is not produced within an elapsed period of 90 seconds. Repeat the test, allowing the timer to operate for periods of 10 plus or minus 1 second and 16 plus 0 minus 1 second before removing the applied potential. No electrical output pulse exceeding 1.5 volts shall be produced.

4.6.1.4 Leakage current.

4.6.1.4.1 Timer interval, 12.5-second. Apply a potential of 32 volts DC to the input of the 12.5-second section, using the tester specified in 4.4.1a. The output voltage shall not exceed 25 millivolts prior to the electrical output pulse.

4.6.1.4.2 Timer interval, 18.5-second. Apply a potential of 32 volts DC to the input of the 18.5-second section, using the tester specified in 4.4.1a. The output voltage shall not exceed 25 millivolts prior to the electrical output pulse.

4.6.1.5 Visual examination. Visual examination shall be made on each timer submitted for delivery to the procuring agency. The timer shall be examined for conformance to the requirements of good workmanship specified in 3.6. Nonconformance is cause for rejection of individual unit.

4.7 Electrical component inspection tests. Samples, selected as specified in Table III of this specification, shall be subjected to the tests and examinations described in 4.7.1.

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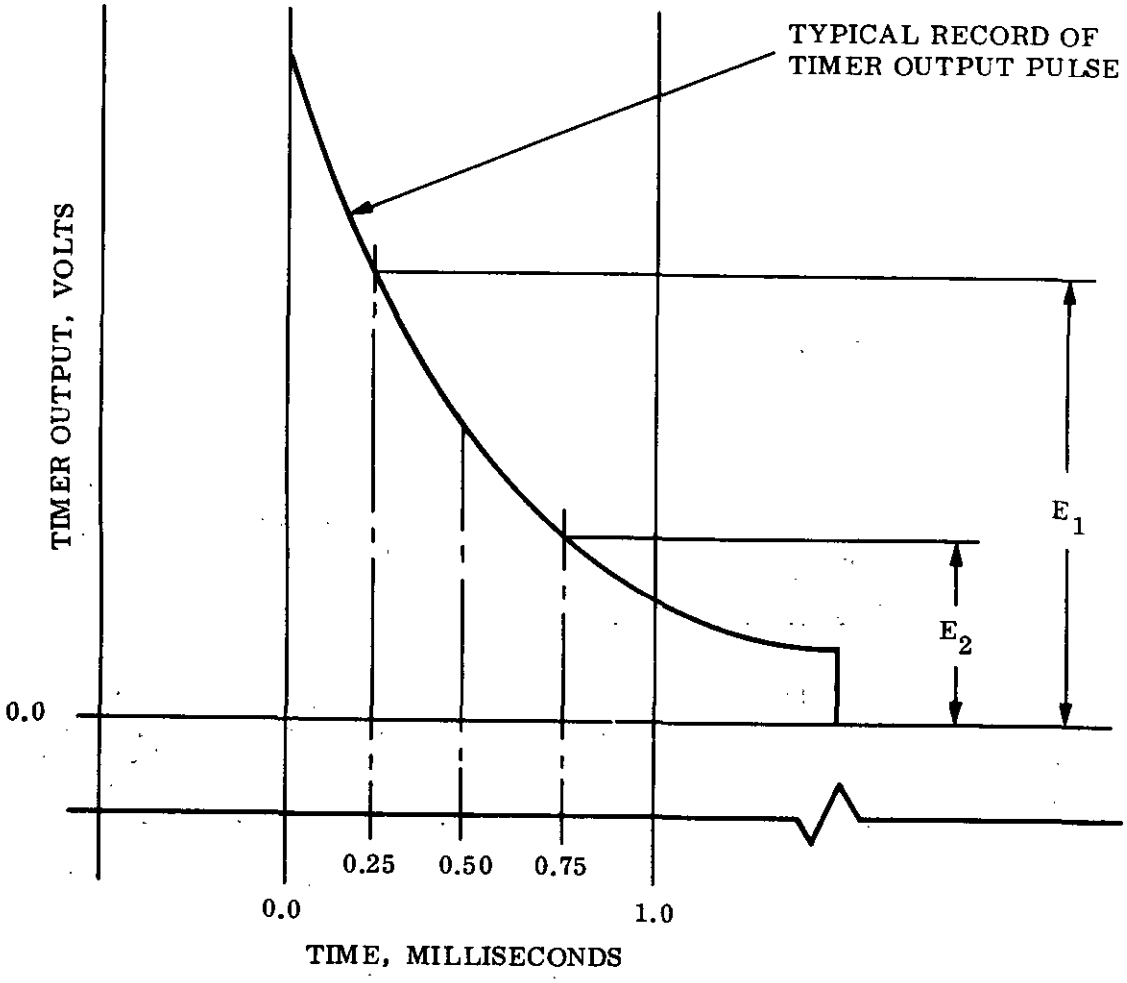


Figure 4. Typical Record of Timer Output Pulse

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Table III. Component Tests, Sampling Plan, and Acceptance Level

Test Paragraph	Test	Sampling Plan	Acceptance Level
4.7.1.1	Capacitor and resistor functional test	*	*
4.7.1.2	Semiconductor thermal shock test	*	*
4.7.1.3	Capacitor, resistor and semiconductor mechanical shock test	**	**

*One-hundred percent inspection; defective units shall be rejected.

**MIL-STD-105 D Special Inspection Level S-3; zero defects accepted; selected components may be resubmitted as allowed in 4.7.1.3.

4.7.1 Test procedure. The compcnents shall be subjected to the test and examinations outlined herein. Unless otherwise specified, inability of the components to pass the tests and examinations shall constitute a failure.

4.7.1.1 Capacitor and resistor functional tests. Tests shall be performed on each capacitor and resistor to determine whether capacitance, resistance, and leakage values are within the limits specified on the detail drawings.

4.7.1.2 Semiconductor thermal shock test. Each semiconductor shall be functionally tested at room temperature to determine conformance with the detail drawings. Then the semiconductor temperature shall be changed from room ambient temperature to minus 65 plus 5 minus 0 degrees F in 1 minute maximum, from minus 65 to plus 160 plus 0 minus 5 degrees F in 2 minutes maximum, and returned to room ambient temperature in 1 minute maximum. The complete temperature cycle shall be accomplished in 4 minutes maximum. All semiconductors shall be temperature-cycled 10 times and again functionally tested to determine conformance with the detail drawings.

4.7.1.3 Capacitor, resistor, and semiconductor mechanical shock test. The component, mounted in a similar manner to that shown on BuWeps Drawings 1660468, 1660469, and 1984195, shall be capable of withstanding one shock of 2400 plus or minus 240g having a pulse shape approximating a one-half sine wave with a duration of 1 plus or minus 0.25 millisecond along each of the mutually perpendicular axes, longitudinal, lateral, and vertical. During and after the shock, the component shall meet the requirements on the detail drawings.

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These shock requirements shall be verified by testing a sample of each lot in accordance with MIL-STD-105 D, special inspection level S-3. No defects shall be permitted in the sample. The sample units shall not be used in the assembly of the timers. Rejected lots of unijunction transistors and silicon-controlled rectifiers cannot be resubmitted. Rejected lots of diodes, resistors, and capacitors as listed in Figure 5 may be resubmitted for qualification by shock testing all components of the lot. The resubmitted components shall be shocked once, with a magnitude of 2430 plus or minus 110g and a pulse shape approximating a one-half sine wave having a duration of 1 plus or minus 0.25 millisecond, in the direction shown in Figure 5. During and after shock, the components shall meet the requirements of the detail drawings. Failure of a resubmitted component to meet these requirements shall result in rejection of that component. The resubmitted components that meet the requirements of the detail drawings then may be used in the assembly of the timers.

4.8 Preservation, packaging, packing, and marking. The timer shall be examined for conformance to the prescribed preservation, packaging, packing, and marking requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified. (See 6.2b and 6.4.)

5.1.1 Level A.

5.1.1.1 Cleaning. The timer shall be cleaned in accordance with Process C-1 of MIL-P-116.

5.1.1.2 Drying. The timer shall be dried in accordance with Procedure D-4 of MIL-P-116.

5.1.1.3 Packaging. The timer shall be packaged in accordance with Method IA-5 of MIL-P-116. The timer shall be cushioned to prevent damage.

5.1.2 Level C. The cleaning, drying, preservation, and packaging shall be in accordance with the supplier's commercial practice, when such practices have been approved by the procuring activity.

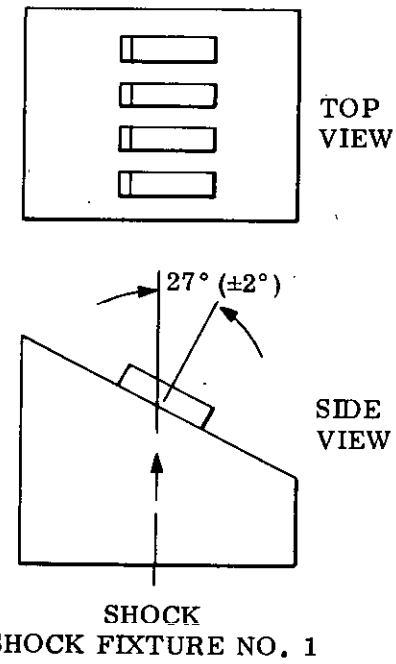
5.2 Packing. Packing shall be Level A or C as specified. (See 6.2c and 6.4.)

5.2.1 Level A. Unless otherwise specified in the contract, order, or requisition, the timer shall be packed in exterior type shipping containers PPP-B-621 or uniform size and shape, with the minimum cube and tare consistent with the protection required. The gross weight of each pack shall be limited to approximately 100 lbs. Containers shall be provided with a case liner, MIL-L-10547, and shall be closed and strapped in accordance with PPP-B-621.

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1. The following components shall be resubmitted for shock tests in Shock Fixture No. 1:

Capacitors	Resistor
1660716	2102987
1660717	



2. The following components shall be resubmitted for shock tests in Shock Fixture No. 2:

Diodes	Resistors
2102988	1660525-1
2102989	1660525-2
	1660525-3
	1660525-4
	1660525-5
	1660525-6

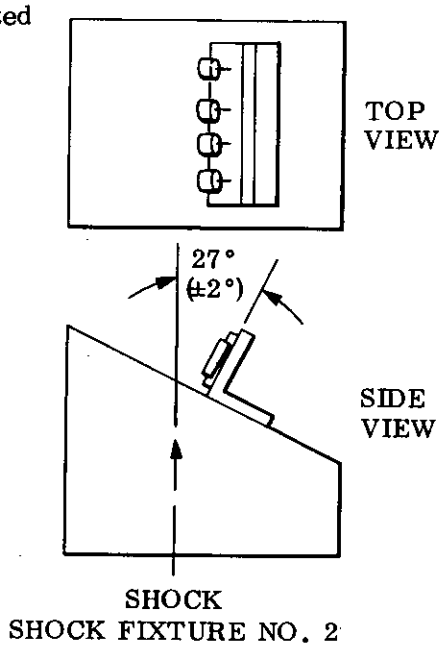


Figure 5. Component Resubmittal Shock Test

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5.2.2 Level C. The timer, packaged as specified in 5.1.2, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use and description.

6.1.1 Intended use. The timer is intended to be used in the fuzing circuit of a depth charge.

6.1.2 Description. The timer is a two-section, electronic device designed to produce separate electrical pulses at predetermined time intervals of 12.5 and 18.5 seconds. Both timing circuits are assembled onto circuit boards and are encapsulated in a polyurethane compound within a metal case with the lead wires passing through a rubber nipple. The timing functions are electrically independent and are measured from the instant of application of voltage.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Required level of preservation and packaging

c. Required level of packing

d. Any exception to this specification

6.2.1 Additional samples. Additional samples required because of failure of any sample(s) to pass the prescribed inspections shall be provided and inspected at the expense of the supplier.

6.3 Definitions. For the purpose of this specification, the following definitions have been established.

6.3.1 Unit condition.

6.3.1.1 Operating. Operating is the condition of the timer during missile operation.

6.3.1.2 Standby. Standby is the condition of the timer when aboard ship and installed in the end item.

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6.3.2 Electrical component inspection. Electrical component inspection are those tests performed on the electrical and electronic components of the timer prior to assembly.

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as specified in FED-STD-102.

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